SETUP LAB · FRC JAVA CRASH COURSE

Setup lab: tools and robot

WPILib, Game Tools, roboRIO, radios and SPARK MAX

WPILib 2026 · Java · Command-based

SETUP

What has to be done

1. **Install WPILib**
   Each laptop.

2. **Install the FRC Game Tools**
   Each laptop.

3. **Image the roboRIO**

4. **Program the radios**
   Both radios.

5. **Set up the motor controllers**
   Every SPARK MAX.

JOB 1: INSTALL WPILIB

Get the WPILib installer

1. **Download the Windows .iso**
   Version 2026.2.1, from the link or the install USB stick.

2. **Right-click it and choose Mount**

3. **Run WPILibInstaller.exe**
   If Windows warns you, click More info, then Run anyway.

WPILib install guide

docs.wpilib.org: WPILib Installation Guide

JOB 1: INSTALL WPILIB

Run the WPILib installer

1. Press **Start**, then choose **Everything**

2. Choose **Install for this User**

3. Choose **Download for this computer only**
   From a USB stick, choose **Select existing VS Code archive**.

4. Press **Finish**, then open **2026 WPILib VS Code**

JOB 2: INSTALL GAME TOOLS

Install the FRC Game Tools

1. **Find FRC Game Tools 2026 on the NI site**
   Log in with an NI account.

2. **Click Download**
   From a USB stick, mount the .iso and run install.exe.

3. **Click Yes and Next through the screens**
   Accept the license. On Fast Startup, change nothing and click Next.

4. **Reboot if asked**
   Then find **FRC Driver Station** on the desktop.

Game Tools install guide

docs.wpilib.org: Installing the FRC Game Tools

JOB 3: IMAGE THE ROBORIO

Find out which roboRIO you have

Look for a microSD card slot.

No card slot: roboRIO 1

Use the next slide, and skip the one after.

Card slot: roboRIO 2

Skip the next slide.

JOB 3: IMAGE THE ROBORIO

Image a roboRIO 1

1. **Robot on, then connect the USB cable**
   The printer-style cable goes in the USB Device port.

2. **Open the roboRIO Imaging Tool**
   Nothing found? Try Run as Administrator.

3. **Click Format Target**
   Enter your team number, pick the latest image, click **Reformat**.

4. **Wait for it to finish**
   Click OK, then Close.

5. **Press the Reset button**

roboRIO 1 imaging

docs.wpilib.org: Imaging your roboRIO 1

JOB 3: IMAGE THE ROBORIO

Image a roboRIO 2

1. **Find the image file**
   In the Imaging Tool, click the SD button. Use the newest FRC_roboRIO2 zip.

2. **Put the card in a USB reader on your laptop**
   A card of 2 GB or more is recommended.

3. **Write it with balenaEtcher or Raspberry Pi Imager**
   If Windows offers to format the card, say no.

4. **Power off, insert the card, power on**

5. **Set the team number**
   Imaging Tool: **Edit Startup Settings**, Team Number, **Apply**.

roboRIO 2 imaging

docs.wpilib.org: Imaging your roboRIO 2

JOB 4: PROGRAM THE RADIOS

Connect to the radio

1. **Plug Ethernet into the port marked DS**
   The other end goes in your laptop.

2. **Power the radio**
   Use the PoE adapter or 12 V.

3. **Turn off your laptop’s Wi-Fi**

4. **Open http://radio.local/**
   Will not load? Run the Vivid-Hosting Network Assistant Tool.

VH-109 quick start

Vivid-Hosting: Quick Start Guide

JOB 4: PROGRAM THE RADIOS

Update the radio firmware

1. **Get firmware 2.0.1**
   Radios on 1.2.x or newer use the file with FROM_1_2_X in its name.

2. **Paste its SHA-256 key into the Checksum box**

3. **Click Upload**
   Keep the power on until it finishes.

4. **Wait for the PWR light to go solid**
   The **SYS** light then blinks slowly.

5. **Repeat on the second radio**

Radio firmware

Vivid-Hosting: Firmware Releases

JOB 4: PROGRAM THE RADIOS

Set up the two radios

1. **First radio: choose Robot Radio Mode**
   Enter your team number and 6 GHz key, then click Configure.

2. **Second radio: choose Access Point Mode**
   Use the same team number and key.

3. **Power the access point from the wall**
   Not from a battery.

4. **Write down the 6 GHz key**
   Keep it with the robot.

JOB 5: SET UP THE MOTOR CONTROLLERS

Update a SPARK MAX

1. **Install REV Hardware Client 2**
   Not the old REV Hardware Client.

2. **Turn the robot off**

3. **Plug USB into one SPARK MAX**

4. **Update the firmware**
   Pick the newest 26.x, even if a newer version is listed.

REV Hardware Client 2

docs.revrobotics.com: REV Hardware Client 2

JOB 5: SET UP THE MOTOR CONTROLLERS

Give each SPARK MAX its settings

1. **Give it a CAN ID from 1 to 63**
   Every SPARK MAX gets a different ID.

2. **Set the motor type**
   Brushed for CIM, REV NEO Brushless for NEO. Brushed mode can ruin a NEO.

3. Click **Persist Parameters**

4. **Label it and write down its ID, then do the next one**

5. **Check that they all show up**
   Power on, USB in any one, then look under **CAN Bus**.

REFERENCE

Official docs



Zero to Robot

docs.wpilib.org: Introduction



Robot wiring

docs.wpilib.org: Robot Wiring



Offline installs

docs.wpilib.org: Offline Preparation



Status lights

docs.wpilib.org: Status Light Reference